OPEN PROJECTS SUMMER 2025
FINANCE CLUB

CREDIT CARD DEFAULT PREDICTION

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A REPORT ON CREDIT CARD BEHAVIOR SCORE PREDICTION USING CLASSIFICATION AND RISK BASED TECHNIQUES

Overview of Approach and Modeling Strategy

The objective of this project is to predict whether a credit card customer will default in the next month, enabling Bank A to proactively mitigate financial risk. The dataset contains 30,000+ labeled customer records with behavioral and demographic information.

Data Cleaning and Preprocessing

To prepare the training dataset for modeling, the following data cleaning and preprocessing steps were conducted on the training dataset:

1. Handling Missing Values

- The dataset was examined for missing values using isnull().sum() to ensure data completeness.
- The age column was found to contain some missing values.
- As age is a numerical and often skewed variable, missing values were imputed using the median, which is more robust to outliers than the mean.

2. Addressing Undefined Categories

- Categorical variables such as education and marriage were analyzed using value_counts() to identify undefined or rare categories:
  - In the education column, values like 0, 5, and 6 were not well defined.
  - In the marriage column, the value 0 was also undefined.
- These values were grouped into the "Others" category to avoid noise while preserving potential predictive power.

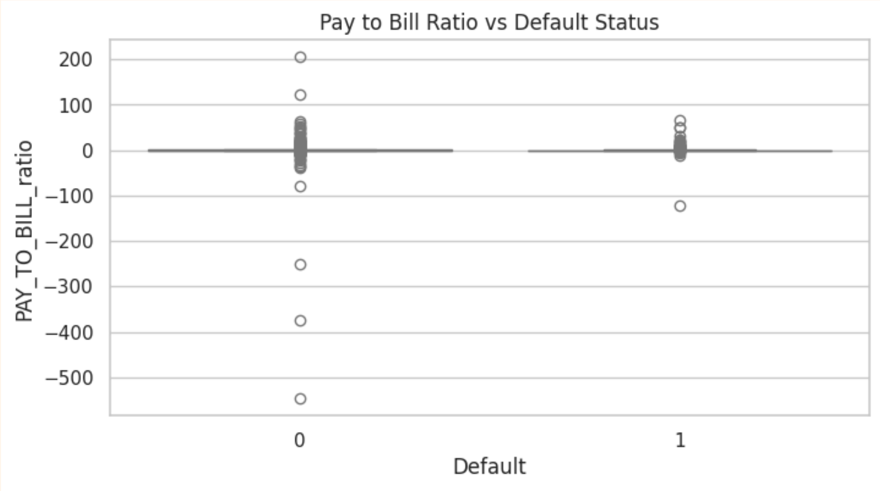
3. Encoding Categorical Variables

- To make the dataset suitable for machine learning models, especially tree-based ones and Logistic Regression, we applied One-Hot Encoding to the categorical variables education and marriage.

- This step helped convert categorical labels into binary columns, allowing algorithms to interpret them appropriately without assuming any ordinal relationships.
- Specifically, the education and marriage columns were encoded into separate binary columns such as education_2, education_3, education_4, marriage_2, and marriage_3.

4. Correction of Data Quality Issues

- Through boxplot visualizations like the one below (Pay to Bill Ratio vs Default Status), we discovered that some rows had implausibly negative values for AVG_Bill_amt, which should represent the average billed amount and thus be non-negative.



- These errors propagated into the derived feature PAY_TO_BILL_ratio, making it highly skewed and incorrect in those rows.
- To correct this:
  - We manually recalculated AVG_Bill_amt using the six monthly BILL_AMT columns for those rows.
  - We then recomputed the PAY_TO_BILL_ratio accordingly to ensure this important financial feature was accurate and trustworthy.

5. Feature Engineering

4. Ensuring Consistency Across Datasets

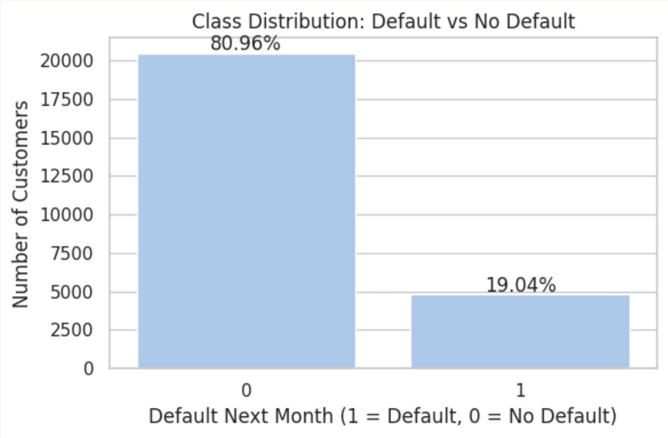
- All feature transformations were applied consistently to both the training and validation datasets to avoid data leakage and ensure reliable model evaluation.

Exploratory Data Analysis Findings and Visualizations

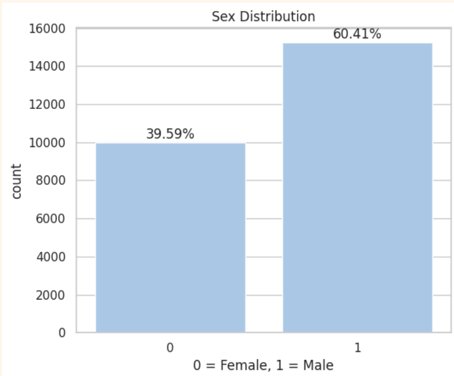
The primary goal of the EDA was to understand the structure, quality, and distribution of the data, and to identify patterns associated with credit default behavior. Key insights and visualizations are outlined below:

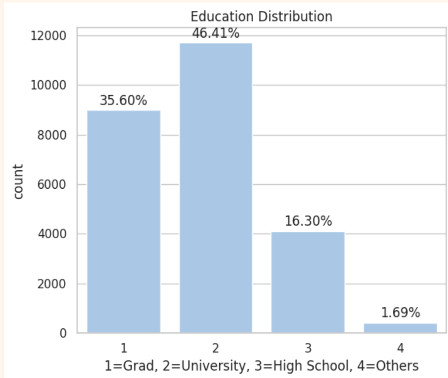
1. Class Imbalance

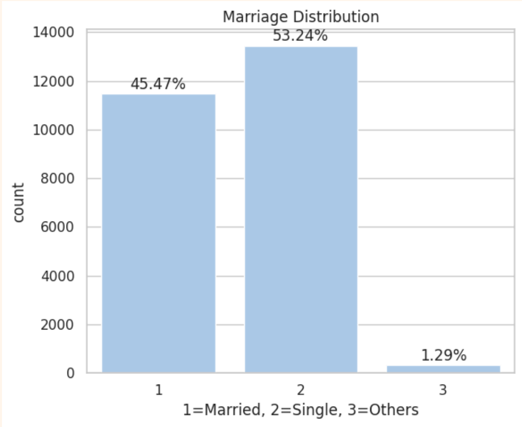
- The target variable next_month_default was found to be highly imbalanced, with a much smaller proportion of defaulters (~19%) compared to non-defaulters (~81%). This imbalance prompted the use of specialized techniques such as **SMOTE** and **class weighting** during model training to avoid bias toward the majority class.
- Models tend to favor the majority class (0) by default, often predicting "No Default" for most customers to get high accuracy.
- Accuracy becomes misleading — predicting all 0s gives ~81% accuracy, but fails to catch any actual defaults.
- Recall on class 1 (defaults) becomes the key metric for risk assessment.

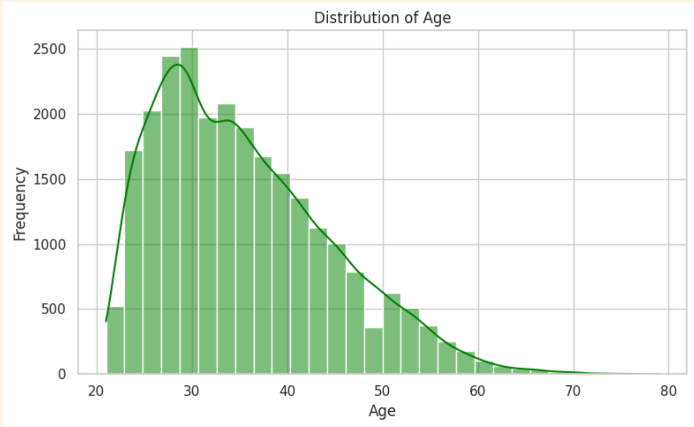


2. Demographic Class Distributions (Univariate Analysis)



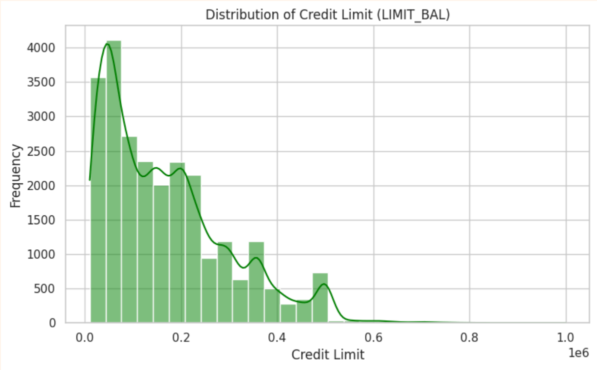


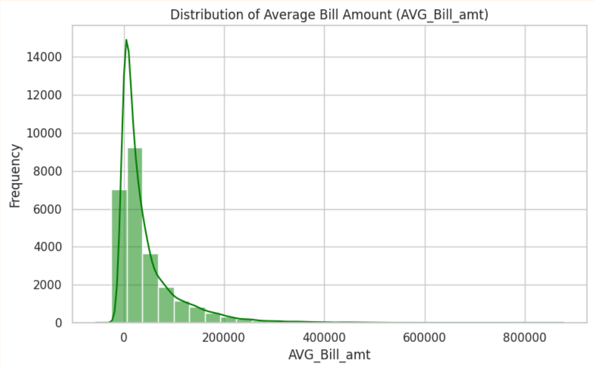




- The chart shows that 60.41% of customers are male and 39.59% are female, indicating a gender imbalance in the dataset that may affect model fairness. No action is needed unless gender significantly impacts predictions or fairness is a concern which is not so in our case.
- Most customers are university-educated (46.41%), followed by graduates (35.60%) and high school-educated individuals (16.30%).
- The majority of customers are single (53.24%), followed by married individuals (45.47%), with a very small portion categorized as others (1.29%).
- Most customers are in their late 20s to early 30s, with the age distribution skewed toward younger individuals.

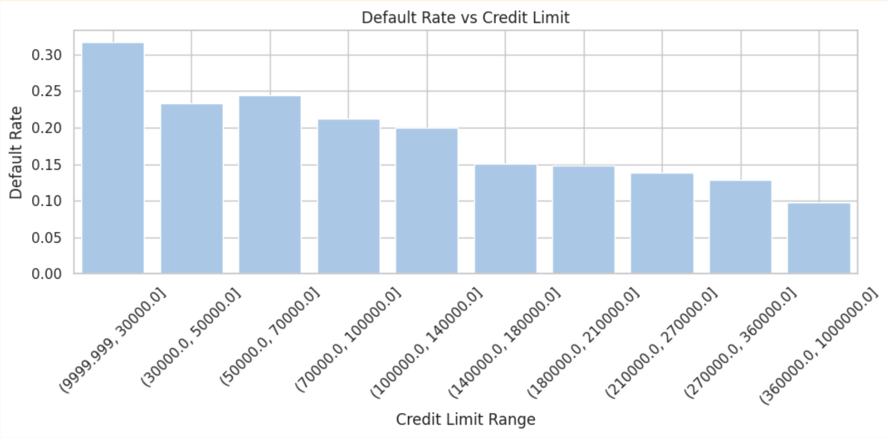
3. Univariate Analysis of Financial Variables:





4. Bivariate Analysis:

4.1 Credit Limit and Default

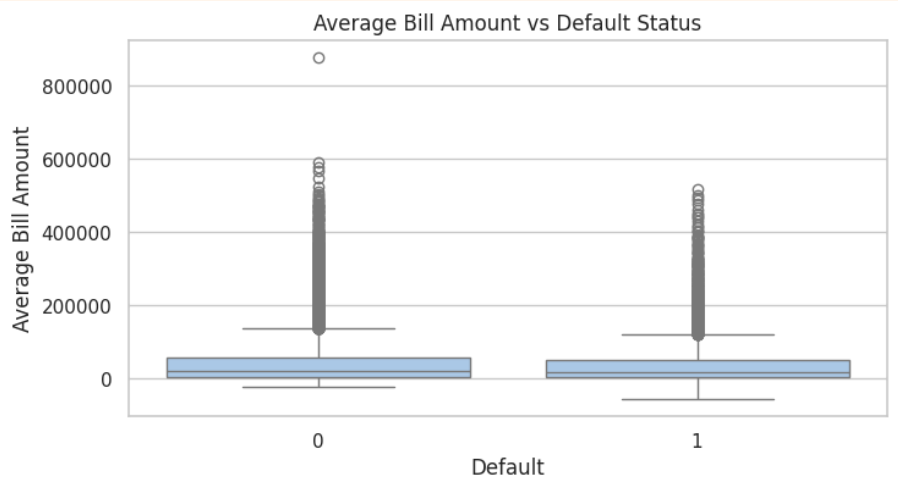


Default rate clearly decreases as the credit limit increases.

Customers with low credit limits (under 30,000) have a default rate of over 30%, while those with limits above 360,000 have default rates below 10%. This indicates a negative correlation between credit limit and default risk.

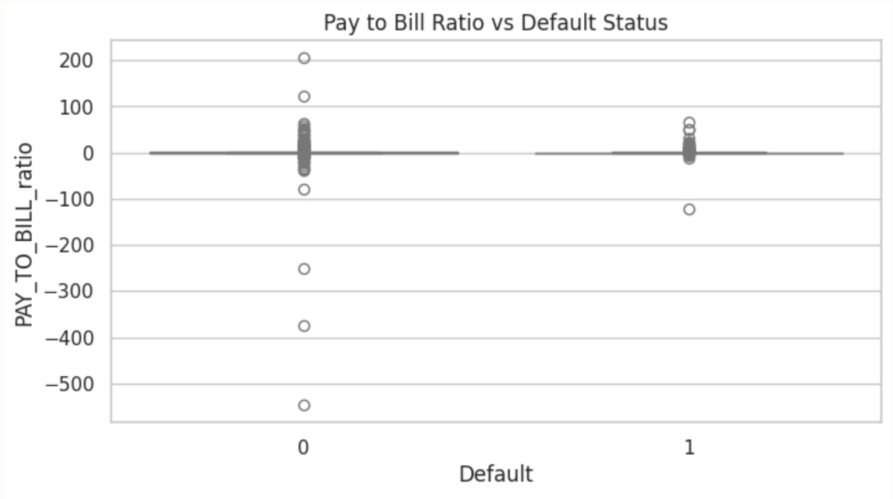
Therefore, customers with lower credit limits are significantly more likely to default, indicating that credit limit can be a strong predictor of credit risk.

4.2 Average Bill Amount and Default



Almost no variation in average bill amounts of defaulters and no defaulters.

4.3 Pay to Bill Ratio and Default



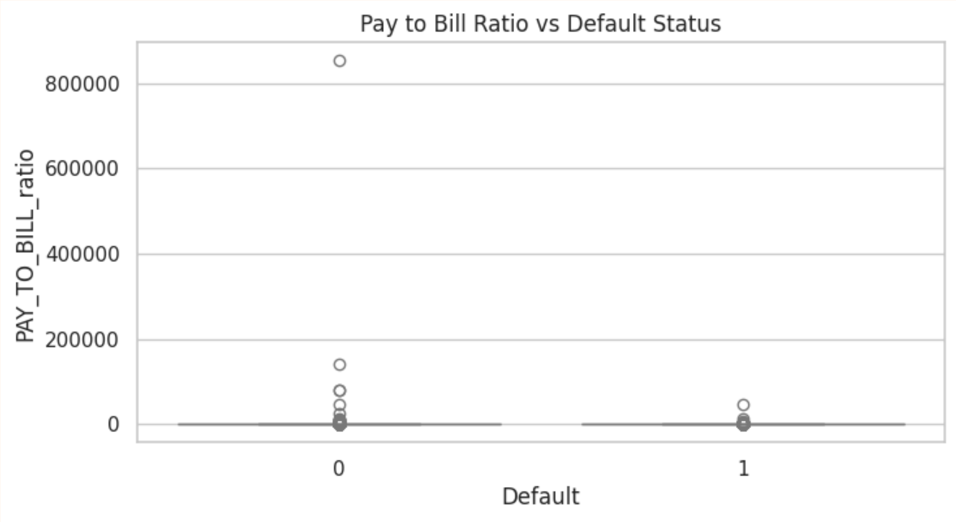
The Pay-to-Bill ratio distribution was highly skewed with extreme outliers in both default and non-default groups. It was observed that there were many negative values of the pay-to-bill ratio in the dataset, which was either because of negative average pay amount or negative average bill amount. Possible reasons:

1. If a user returns items or receives a rebate, the credit card statement might show a negative bill for that month leading to a negative bill amount.

2. Errors during data recording or cleaning might mistakenly assign negative values to bill amounts.

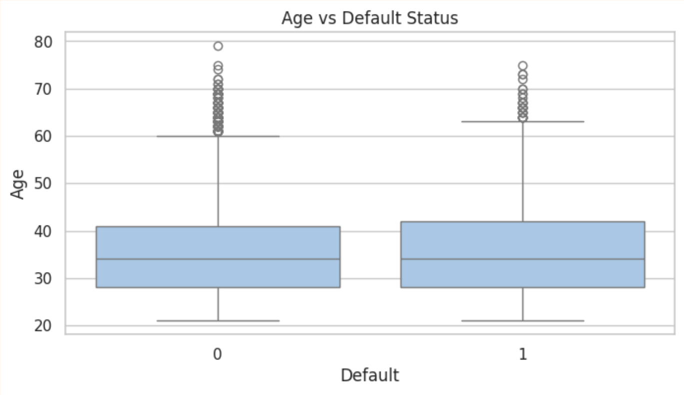
3. If the user overpaid last month, the statement might reflect a negative balance as a carryover.

We next looked closely to observe our dataset for what might be the actual reason behind these anomalies which turned out to be errors in the calculation of AVG_Bill_amt column which was rectified manually as explained in the data cleaning and preprocessing section. New distribution as follows:

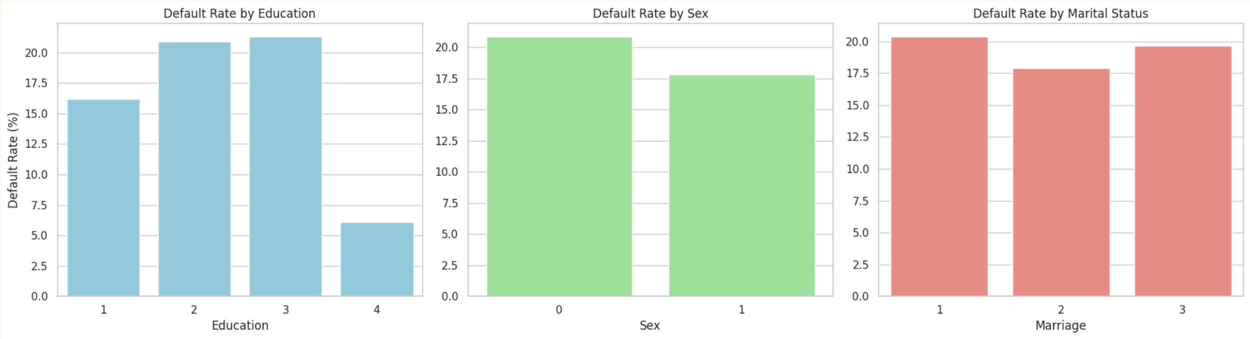


4.4. Demographics and Default Risk

- No major correlation between age and default status could be observed.



- Lower formal education (especially high school or university) is associated with a higher likelihood of default. This might reflect financial literacy, income stability, or access to credit.



- Males have a higher default rate ~21% than females ~18%. Gender may play a role in credit behavior, possibly influenced by spending patterns, income distribution, or risk appetite.
- Marital responsibilities may affect financial commitments or stress levels. Could also be a function of income-to-expense ratio or dependents.

5. Results from the Correlation Heatmap plotted for all relevant columns

1. Bill_amt1 to Bill_amt6 and AVG_Bill_amt are very highly correlated with each other (0.8–0.97). We can reduce dimensionality here by keeping just the average.

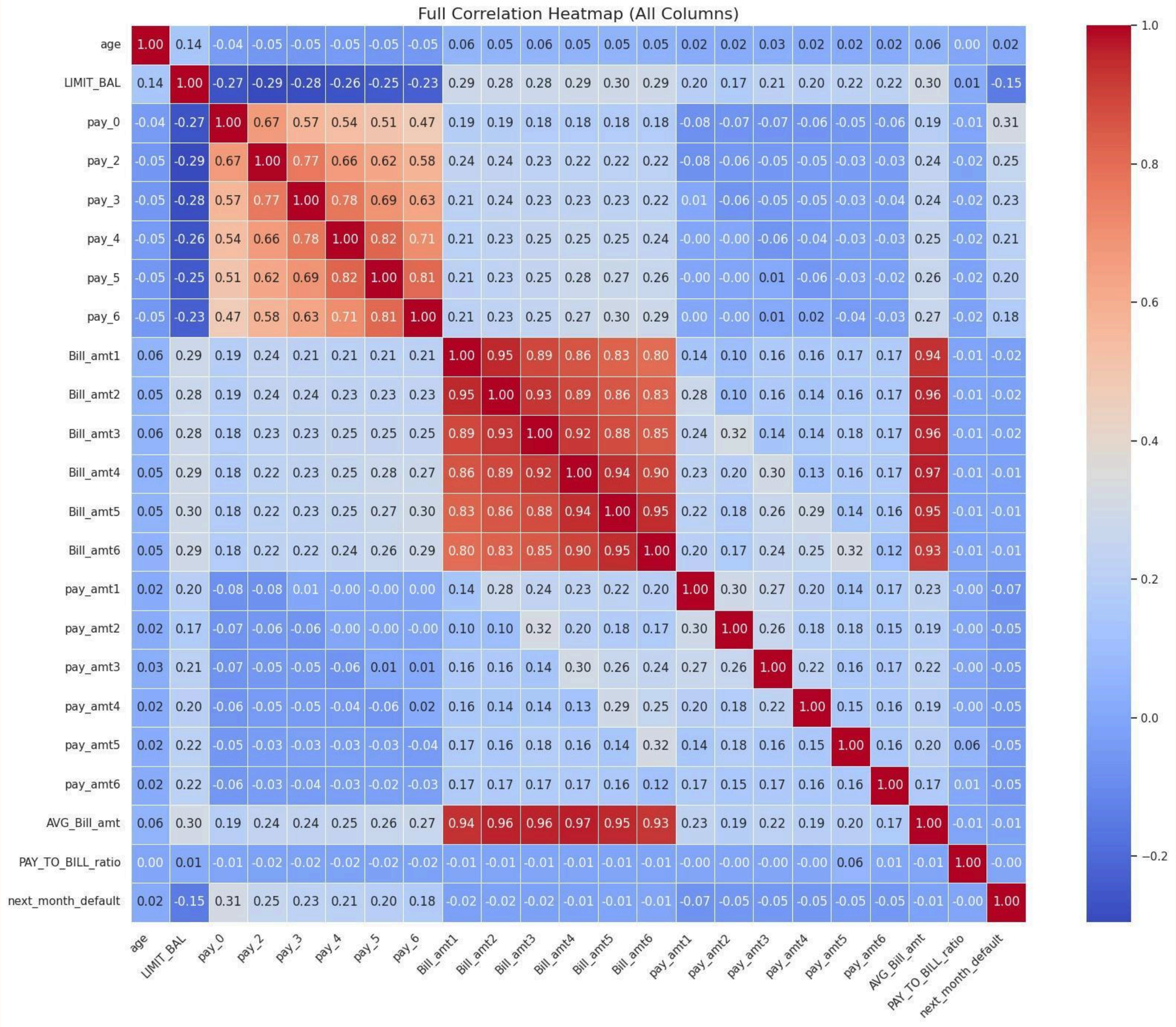
2. PAY features (pay_0 to pay_6): Highly correlated among themselves (0.5–0.77). Indicates users tend to repeat payment behavior (delayed once, delayed again). Correlated positively with next_month_default (~0.18 to 0.31).

3. PAY_TO_BILL_ratio: Very low correlation with next_month_default (-0.01). May need feature engineering or non-linear models to capture its value.

4. Payment Amounts (pay_amt1 to pay_amt6): Slightly negatively correlated with default. Larger payments → less chance of default. But correlation is weak (around -0.01 to -0.06), meaning not strong linear influence

5. LIMIT_BAL: Small negative correlation with default: -0.15. Higher limits → slightly lower chance of default. Likely tied to customer creditworthiness.

6. age: Correlation with default: -0.01. Very low — not a linear indicator of default

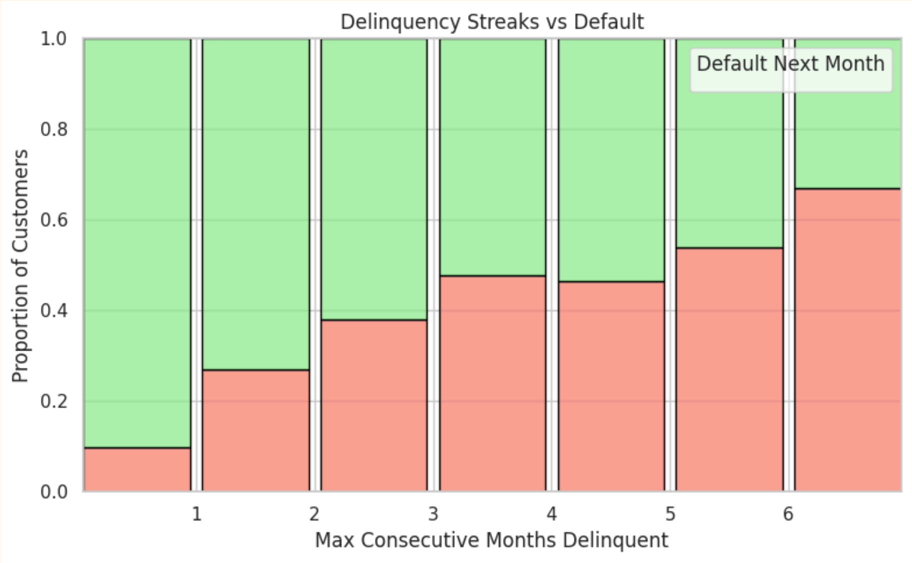


- Variables such as PAY_0, PAY_2, etc., showed a clear association with default. Higher delay codes (e.g., 2, 3, 4, 5) were significantly more common among defaulters.
- This motivated the creation of aggregate delay features like:
    - avg_delay: Average delay across months.
    - count_late: Number of months with late payment.
    - max_delinquency_streak: Longest consecutive delay stretch.

Analyzing Behavioral Trends through Feature Engineering + Financial Insights

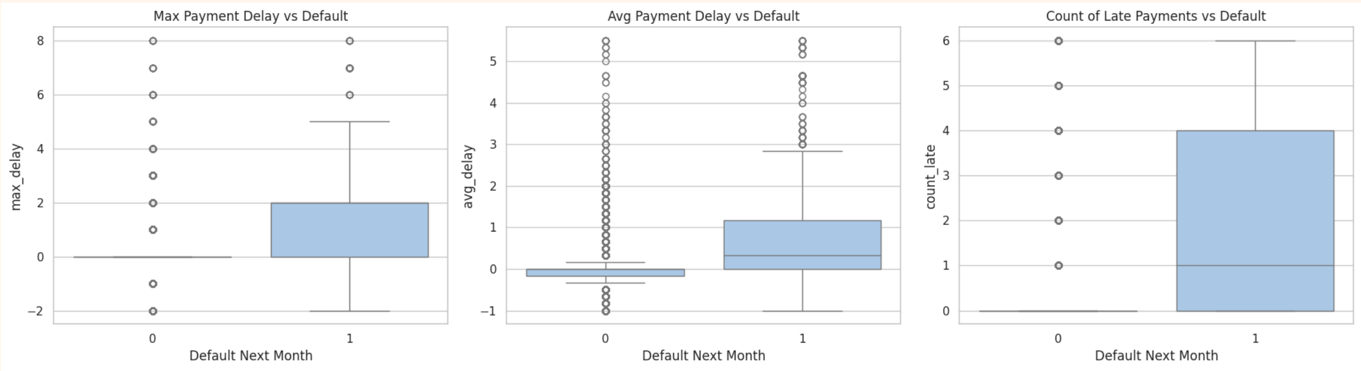
We added several new features based on domain intuition and temporal payment behavior:

- **total_bill_amt and total_pay_amt**: Aggregate billed and paid amounts respectively.
- **Max_delinquency_streak:** refers to the longest consecutive period (in months) a customer had delayed payments without returning to on-time repayment.



The graph highlights a clear behavioral trend: as the maximum number of consecutive delinquent months increases, the probability of default rises significantly. This feature, max_delinquency_streak, captures the persistence of missed payments and is more informative than isolated delays. Customers with longer delinquency streaks (4 months or more) show a much higher likelihood of default, indicating chronic financial distress.

- **max_delay, avg_delay, count_late**: Indicators of chronic or repeated delays.



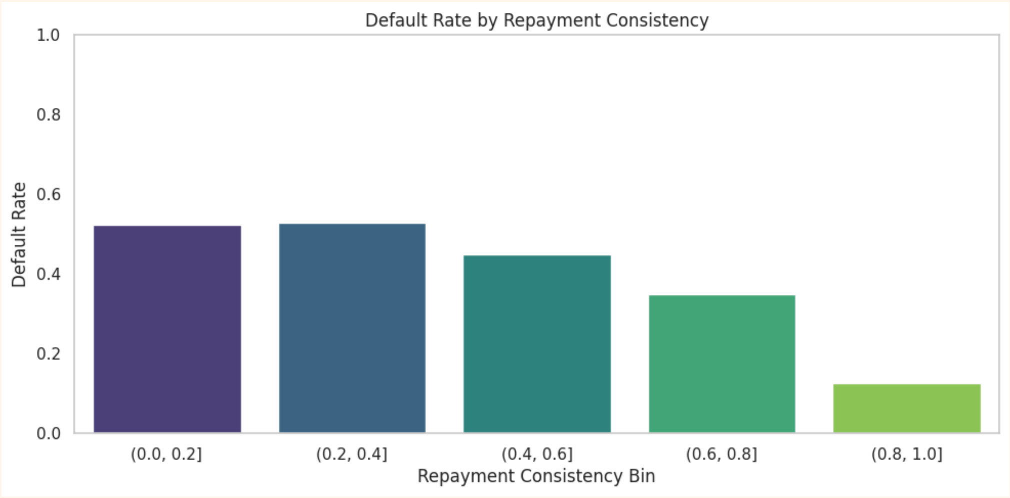
In the first plot, Max Payment Delay vs Default, defaulters show a significantly higher spread and median delay, indicating that individuals with even a single severe delay are more likely to default.

The second plot, Avg Payment Delay vs Default, shows that those who default tend to have a higher average delay across months, reinforcing that chronic lateness is a key predictor.

The third plot, Count of Late Payments vs Default, reveals that defaulters generally have more frequent late payments compared to non-defaulters.

These engineered features — maximum delay, average delay, and count of late payments — effectively capture both the severity and consistency of repayment issues, making them strong indicators of potential credit risk.

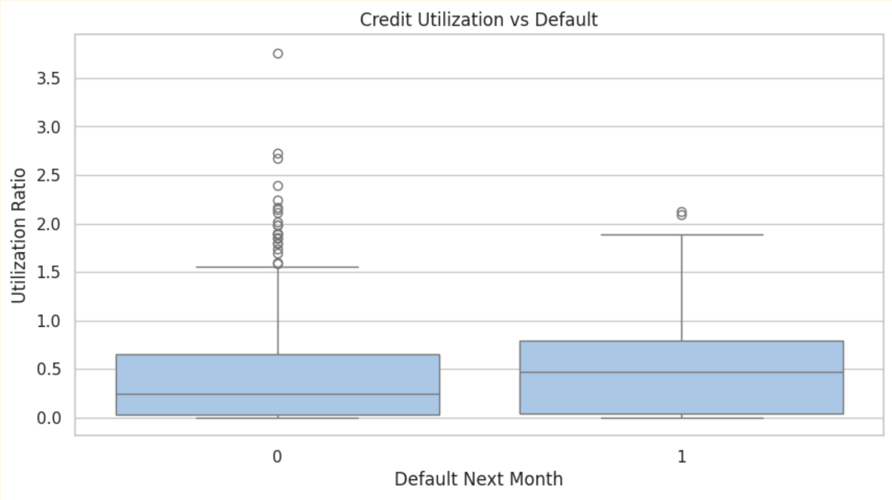
- **repayment_consistency**: This feature quantifies how consistently a customer makes timely or early repayments. It is calculated as the proportion of months (out of 6) in which the repayment status (PAY_x) was on time (0) or early (negative values).



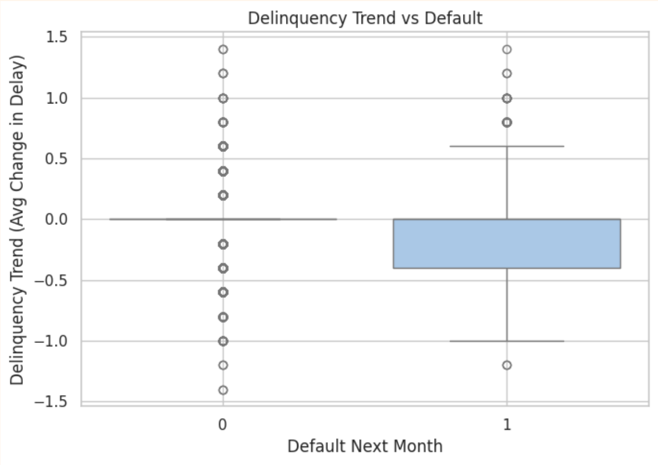
The graph shows that default risk decreases steadily with improved repayment consistency — customers who consistently repay on time (repayment consistency > 0.8) have a default rate below 20%, while those with poor consistency (≤ 0.4) have a default rate exceeding 50%, highlighting **repayment consistency as a strong behavioral predictor of credit risk.**

- **utilization_ratio**: Total Bill Amount / Total Credit Limit, Credit usage level relative to assigned limit.

The boxplot shows that customers who default tend to have higher median credit utilization ratios compared to those who don't, indicating that heavier credit usage is linked to increased default risk



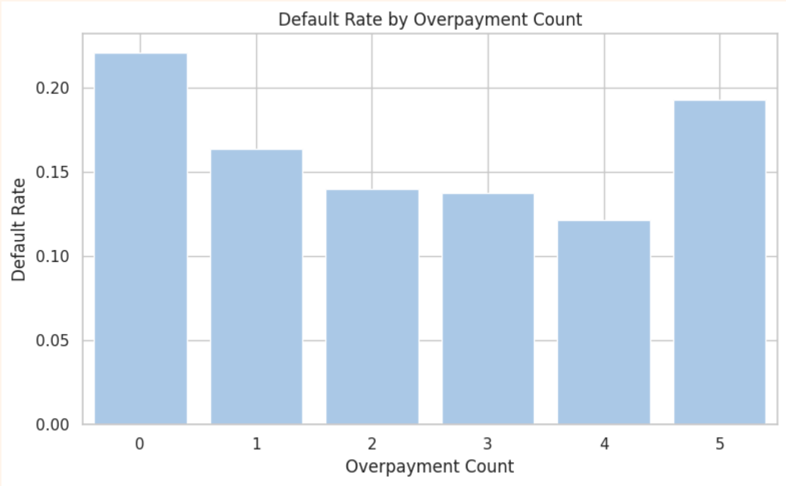
- **delinquency_trend**: Measures whether the user's repayment trend is improving or deteriorating.



This plot shows that defaulters tend to have a negative delinquency trend, meaning their payment status is worsening over time (increasing delays), whereas non-defaulters generally maintain a stable payment pattern with little to no change in delay.

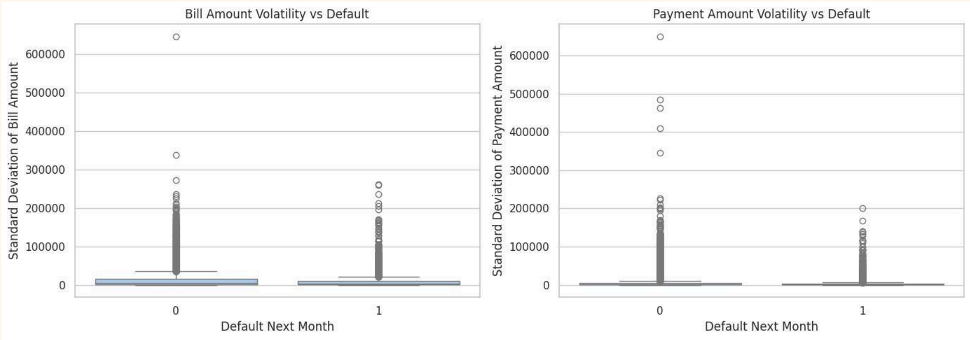
- **overpayment_count**: Counts instances where a customer paid more than they were billed.

This plot shows that customers who defaulted generally had fewer overpayments, suggesting that frequent overpayment behavior is associated with a lower risk of default.



Overpayment frequency can be a useful behavioral signal. A few overpayments suggest proactive debt management, while no overpayments often coincides with higher risk. However, extreme behavior (e.g., always overpaying) doesn't necessarily equate to lower risk and might require further context or investigation.

- **bill_amt_std, pay_amt_std**: Standard deviation in bill and payment amounts.



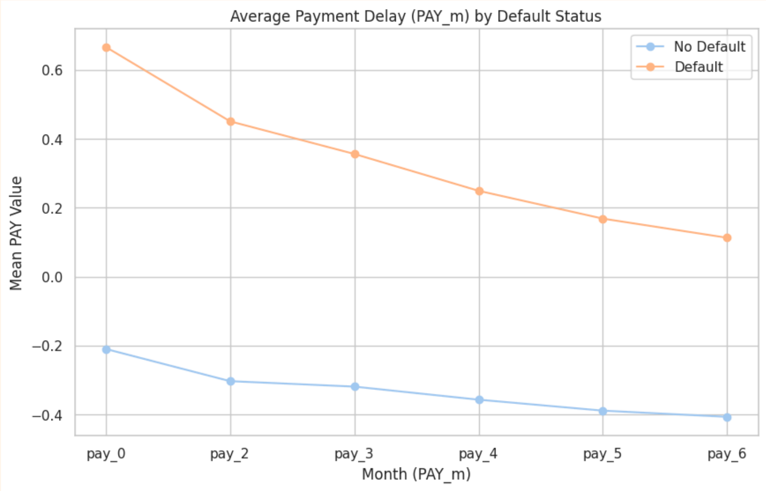
These plots reveal that:

Bill Amount Volatility (left plot): Customers who defaulted tend to have lower variability in their monthly bill amounts compared to non-defaulters. This might indicate consistently high utilization without significant changes in spending, which could signal financial rigidity or stress.

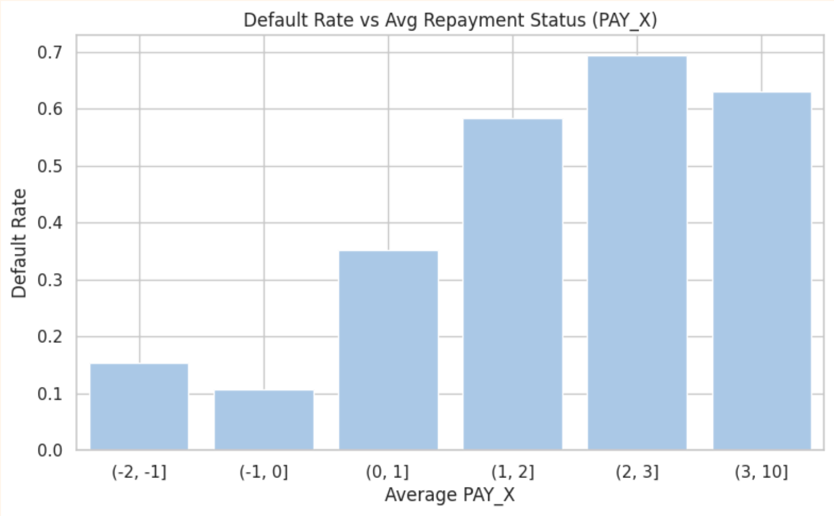
Payment Amount Volatility (right plot): Similarly, defaulters show lower volatility in their payments, suggesting predictable (and often low or minimum) repayments, potentially indicating financial strain or lack of flexibility in managing debt.

Greater volatility in bills and payments may reflect active credit usage and financial engagement, while low volatility paired with default risk might point to financial stagnation or tight constraints — valuable signals when predicting credit risk.

- **avg_PAY_X**: Average of the PAY_X delay statuses over six months.

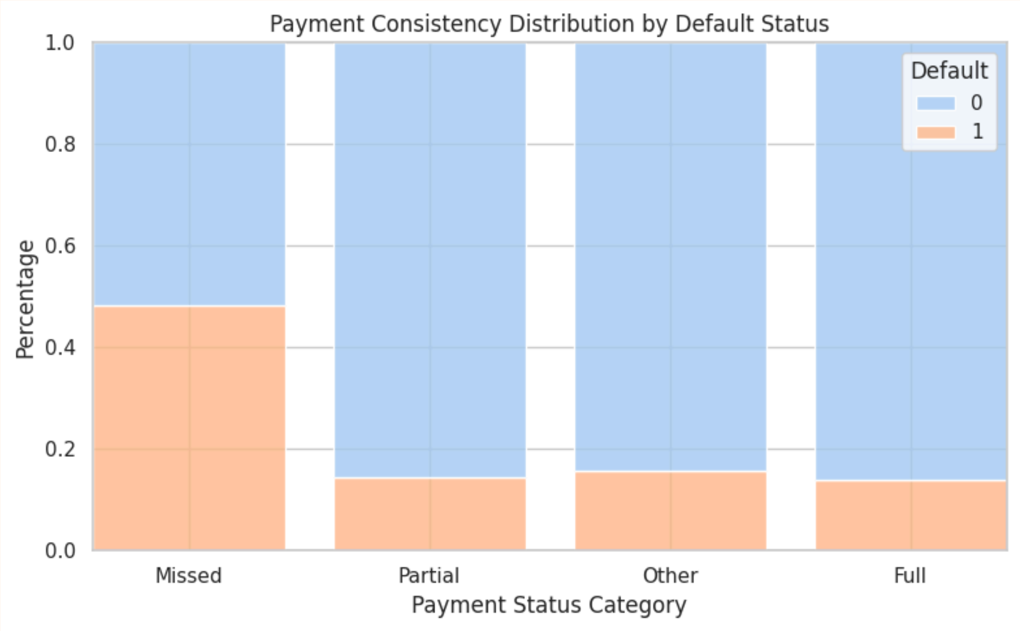


A rising delay trend (especially peaking in recent months like PAY_0) is strongly correlated with default. On the other hand, consistently negative or low PAY values suggest good repayment discipline and lower risk.



The avg_PAY_X feature captures overall repayment timeliness, with higher average values indicating worse payment behavior and a greater risk of default.

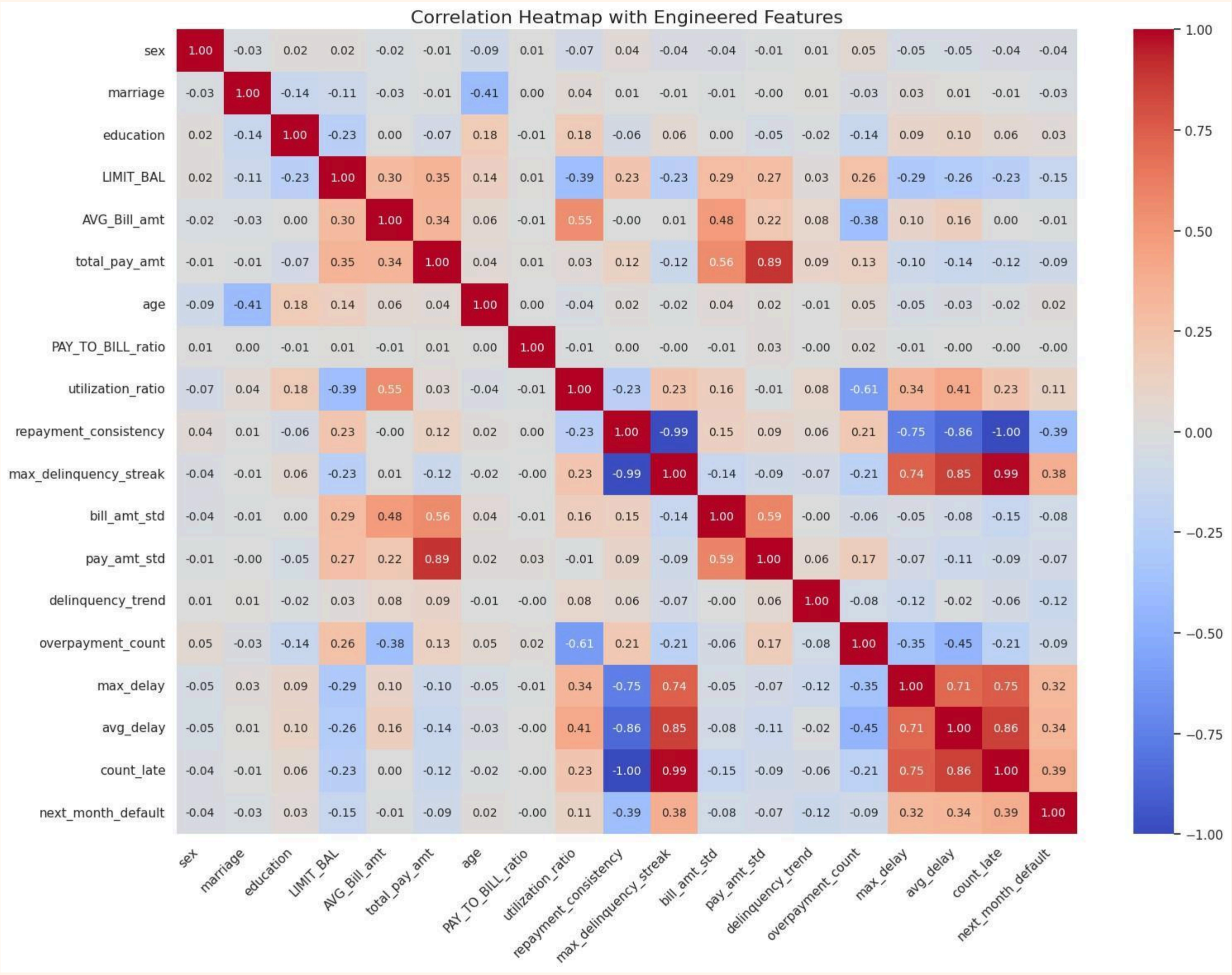
Default risk rises significantly with increasing average delay. Customers with even slight average delays (PAY_X > 0) show a strong uptick in likelihood of default, peaking around 2-month average delay.



This transforms the numeric PAY values into interpretable payment behavior categories, across all months (PAY_0 to PAY_6), and tracks how often customers fall into each category.

This feature captures payment consistency patterns, showing that customers who frequently miss payments are far more likely to default, while those who pay fully or partially on time rarely default.

Correlation Heatmap of Engineered Features



Key Financial Insights on Default Drivers

- **Max Delinquency Streak (+0.39)** and **Count of Late Payments (+0.34)** show the strongest positive correlation with default, indicating that repeated or prolonged delays are strong signs of risk.
- **Repayment Consistency (−0.39)** is highly negatively correlated with default, highlighting the importance of disciplined and regular payments.

- **Average/Max Delay (~+0.30)** also contribute significantly, reflecting how long payments are overdue.
- **Utilization Ratio (+0.11)** suggests overuse of credit may indicate financial stress.
- **Overpayment Count (−0.07)** shows that financially cautious behavior (like paying more than billed) slightly reduces default risk.
- **Demographic features** like sex, education, and marriage show minimal correlation and less predictive value.

Based on the correlation heatmap, we will select only particular features to avoid multicollinearity for logistic regression. Hence we will Drop IDs (non-informative) and avoid duplicate or derived features unless necessary. Therefore

logistic_cols = ['LIMIT_BAL', 'age', 'sex', 'education_2', 'education_3', 'education_4', 'marriage_2', 'marriage_3', 'AVG_Bill_amt', 'bill_amt_std', 'PAY_TO_BILL_ratio', 'pay_amt_std', 'max_delay', 'count_late', 'utilization_ratio', 'delinquency_trend', 'overpayment_count', 'next_month_default']

3.

4. Bill and Payment Amounts

- Features like `AVG_Bill_amt` and `total_pay_amt` were positively skewed with significant outliers.

- Visualizations such as boxplots revealed **erroneous negative values in `AVG_Bill_amt`**, which were cleaned and recalculated.

- A derived feature `PAY_TO_BILL_ratio` (payment as a ratio of bill) was created and corrected to reflect actual financial behavior.

5. Credit Utilization

- The `utilization_ratio` (total billed amount relative to credit limit) showed that **higher utilization was correlated with higher default risk**, likely due to financial stress or overextension.

6. Delinquency Trend

- A new feature, `delinquency_trend`, captured the **direction of repayment behavior**—whether users were improving or worsening in their delay status.

- Defaulters generally had a **flat or worsening trend**, reinforcing its predictive value.

7. Variability Indicators

- Standard deviations of bill (`bill_amt_std`) and payment amounts (`pay_amt_std`) were added to capture **inconsistency in financial behavior**, which often aligns with riskier profiles.

📈 Key Visuals Used:

- **Boxplots**: Used to compare financial behavior between defaulters and non-defaulters (e.g., `PAY_TO_BILL_ratio` vs. `next_month_default`).

- **Correlation Matrix**: Helped identify multicollinearity and the most predictive numeric features.

- **Bar Plots**: Illustrated the distribution of education, marriage status, and repayment history among defaulters.

These preprocessing steps significantly improved data quality, ensured model readiness, and aligned the dataset with best practices for handling categorical variables in machine learning.

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- Engineered domain-informed features (e.g., credit utilization, repayment consistency, delinquency trend)
- Used tree-based classifiers: **Random Forest**, **XGBoost**, and **LightGBM**
- Performed **hyperparameter tuning** and **threshold selection** based on **F2-score**, prioritizing **recall** to minimize missed defaulters
- Evaluated models on a holdout test set and generated final predictions on an unlabeled validation dataset

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📈 EDA Findings and Visualizations

- **Skewed Target Distribution**: ~22% defaulters, ~78% non-defaulters → addressed with **SMOTE oversampling**
- **Correlation Analysis**:
  - Positive correlation: `PAY_X` features (payment delays)
  - Moderate correlation: `BILL_AMT` and `PAY_AMT` (bill and repayment trends)
  - Negative correlation: `repayment_consistency`, `utilization_ratio`
- **Distribution Insights**:
  - Defaulting customers tend to have **high bill amounts but low repayment behavior**
  - High credit utilization and **frequent late payments** were strong indicators of default

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💼 Financial Insights: Key Drivers of Default

Feature	Reason for Inclusion	Financial Insight
`utilization_ratio`	Engineered	High utilization indicates credit stress
`avg_PAY_X`	Engineered	Chronic late payments signal risk
`repayment_consistency`	Engineered	Irregular repayment behavior increases default risk
`max_delay` / `count_late`	Engineered	Captures length and frequency of delay streaks
`AVG_Bill_amt`, `total_pay_amt`	Raw	Reflects burden vs. repayment capacity

`delinquency_trend`	Engineered	Shows whether payment behavior is improving or deteriorating

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🔄 Model Comparison & Final Selection

Model	F2-Score	ROC AUC	Recall (Class 1)	Precision (Class 1)
XGBoost	0.5903	0.7681	0.75	0.32
LightGBM	0.6046	0.7780	0.76	0.33
Random Forest	**0.6093**	**0.7770**	**0.83**	0.30

Final Model: Random Forest, selected for its:

- Highest F2-score (primary metric)
- Excellent recall performance (minimizing false negatives)
- Interpretability for business users

🔢 Evaluation Methodology & Metrics Justification

Prioritized Metric: F2-score ✔

- Heavily weighs **recall** over precision
- In credit risk, **missing a defaulter (false negative)** is costlier than flagging a non-defaulter (false positive)

Metrics on Test Set:

- **Accuracy**: 0.59
- **F1 Score**: 0.44 (Class 1)
- **Recall**: 0.83 (Class 1)
- **Precision**: 0.30 (Class 1)

- **F2 Score**: 0.6093
- **ROC AUC**: 0.7770

🔎 Classification Threshold Selection

- Default threshold (0.5) was **not optimal** for F2
- Tuned threshold using `precision_recall_curve`
- **Selected threshold = 0.67** that **maximized F2-score** on test set
- This increased recall with acceptable trade-off in precision

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💼 Business Implications

- **83% of defaulters** correctly identified ✅
- **Increased proactive risk mitigation**: allows pre-emptive loan denial, credit limit control
- **Higher false positives (1903)**: could impact customer satisfaction, but acceptable given risk appetite

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🌐 Summary of Findings & Key Learnings

- Engineered features like **repayment behavior**, **utilization ratio**, and **payment trends** are powerful predictors
- Tree-based models perform well on imbalanced financial datasets
- Tuning classification threshold is **critical** for business alignment
- F2-score optimization leads to **better risk management** even with trade-offs in precision

Next Steps:

- Monitor model performance over time (data drift, concept drift)
- Integrate with bank's credit pipeline for real-time decisioning
- Consider ensemble stacking or cost-sensitive learning for further optimization